

ELC-112 — Principles of Digital Electronics

Rank	Topic	Importance	Probability / Recurrence
 1	Boolean Algebra + Logic Gates	 10/10	Very High — recurring
 2	Number System & Binary Arithmetic	 10/10	Very High — recurring
 3	K-Map / Boolean Expression Simplification	 10/10	Very High
4	Adder & Subtractor	 9.5/10	Very High — recurring
5	Multiplexer / Demultiplexer	 9.5/10	Very High — recurring
6	Number Codes — BCD, Gray, ASCII	 9/10	Very High
7	SOP / POS / Standard Forms	 9/10	High
8	Universal & Derived Gates	 8.5/10	High
9	BCD to 7-Segment Display	 8/10	Medium-High
10	Encoder / Decoder	 8/10	Medium-High
11	IC 7447 / IC 74138	 7/10	Medium — short-note type
12	Parity Bit / Error Detection	 7/10	Medium — short-note type

1. Boolean Algebra + Logic Gates — 10/10

Exact questions

- "Using rules of Boolean algebra simplify $M = XYZ + XYZ + XYZ + XYZ$." ([Scribd](#))
- "State and prove De-morgan's Theorem." ([Studocu](#))
- "Draw the logic symbol and truth table of AND & OR gate." ([Studocu](#))
- "What is meant by derived gate? Draw the symbol of any two derived gates with their Boolean expression." ([Studocu](#))
- "Write a short note on universal gates." ([Scribd](#))

Prepare these questions

- Define Boolean algebra.
- State Boolean algebra laws.
- State and prove De Morgan's theorems.
- Draw symbols and truth tables of AND, OR and NOT.
- Explain NAND and NOR gates.
- Explain universal gates.
- Explain XOR and XNOR gates.
- Simplify Boolean expressions using Boolean laws.
- Implement basic gates using NAND/NOR.
- Identify derived gates.

 **Priority: MUST DO**

2. Number System & Binary Arithmetic — 10/10

This topic is repeatedly used for **direct numerical marks**, so it is highly scoring.

Exact questions

- "Give rules for binary addition of two bits. Perform $(1100.010)_2 + (10.1110) + (1010)_2$." ([Scribd](#))
- "Give the Rules for two bit Binary Addition and Perform $(11010)_2 + (10101)_2 = (?)_2$." ([Studocu](#))
- "Convert $(45)_{10} - (25)_{10} = (?)_2$ using 2's complement." ([Studocu](#))
- "What is the base of Decimal number system." ([Studocu](#))
- "Write a short note on Hexadecimal number system." ([Scribd](#))

Prepare these questions

- Binary $\rightarrow$ Decimal.
- Decimal $\rightarrow$ Binary.
- Binary $\rightarrow$ Octal.
- Binary $\rightarrow$ Hexadecimal.
- Hexadecimal $\rightarrow$ Binary.
- Binary addition.
- Binary subtraction.
- 1's complement.
- 2's complement.
- Subtraction using 2's complement.
- Number-system bases.
- Hexadecimal number system.

 **Priority: MUST DO**

3. K-Map / Boolean Simplification — 10/10

This is one of the topics where **practice matters more than memorization**.

Exact question

- "Simplify the following expression using K map." ([Scribd](#))

The 2019 paper gives a Boolean expression and explicitly asks for K-map simplification. ([Scribd](#))

Prepare these

- What is a Karnaugh map?
- 2-variable K-map.
- 3-variable K-map.
- 4-variable K-map.
- Simplification using SOP.
- Simplification using POS.
- Don't-care conditions.
- Grouping of 1, 2, 4, 8 cells.
- Derive simplified Boolean expression from K-map.
- Draw logic circuit from the simplified expression.

 **Priority: MUST DO**

4. Adder & Subtractor — 9.5/10

This topic has a very strong recurrence pattern.

Exact questions

- "With neat logic diagram explain the working of 4 bit universal adder subtractor." ([Scribd](#))
- "Draw the logic diagram for 3 bit adder and write its truth table." ([Scribd](#))
- "Draw and explain 4-bit Universal Adder and Subtractor." ([Studocu](#))
- "Draw logic diagram of half subtractor." ([Modern College](#))
- "Draw the truth table of half adder." ([Modern College](#))

Prepare these

- Half adder.
- Full adder.
- Half subtractor.
- Full subtractor.
- 4-bit parallel adder.
- 4-bit universal adder/subtractor.
- Truth table of half adder.
- Truth table of full adder.
- Logic diagram of adders.
- Logic diagram of subtractors.

 **Priority: MUST DO**

5. Multiplexer & Demultiplexer — 9.5/10

This is another topic I would **not skip**.

Exact questions

- "For a multiplexer with 60 inputs, find out the number of control lines." ([Scribd](#))
- "How many select lines are required to design 1:8 Demultiplexer." ([Studocu](#))
- "Draw and explain the logic diagram 1:4 Demultiplexer." ([Studocu](#))
- "Give any 2 uses of De-multiplexer." ([Modern College](#))

Prepare these

- Define multiplexer.
- Define demultiplexer.
- Explain 2:1 multiplexer.
- Explain 4:1 multiplexer.
- Explain 8:1 multiplexer.
- Explain 1:4 demultiplexer.

- Explain 1:8 demultiplexer.
- Calculate number of select lines.
- Applications of multiplexer.
- Applications of demultiplexer.

 **Priority: MUST DO**

6. Number Codes — BCD, Gray, ASCII — 9/10

This is particularly important because the university repeatedly asks **conversion-based questions**.

Exact questions

- "Convert $(1011101)_2 = (?)$ Gray." ([Scribd](#))
- "Convert $(110101)_2 = (?)$ BCD." ([Scribd](#))
- "Convert $(23)_{10} = (?)$ BCD." ([Studocu](#))
- "Full form of ASCII is ____." ([Scribd](#))
- "What is a Gray code? Where is it used?" ([Scribd](#))
- "Write a short note on BCD code. Convert $(125)_{10}$ to BCD." ([Modern College](#))

Prepare these

- Define BCD.
- Decimal $\rightarrow$ BCD.
- Define Gray code.
- Binary $\rightarrow$ Gray.
- Gray $\rightarrow$ Binary.
- Define ASCII.
- Explain applications of Gray code.
- Difference between BCD and Gray code.
- Weighted vs non-weighted codes.

 **Priority: MUST DO**

7. SOP / POS / Standard Forms — 9/10

Exact questions

- "Convert the following expression into standard POS form." ([Scribd](#))
- "Convert given SOP equation to standard SOP: $AB + BC + AC$." ([Studocu](#))

Prepare these

- Define SOP.
- Define POS.
- Standard SOP.
- Standard POS.
- Convert SOP to standard SOP.
- Convert POS to standard POS.
- Minterms.
- Maxterms.
- Canonical forms.
- Convert Boolean expression into logic circuit.

 **Priority: Very High**

8. Universal & Derived Gates — 8.5/10

Exact questions

- "Write a short note on universal gates." ([Scribd](#))
- "What is meant by derived gate? Draw the symbol of any two derived gates with their Boolean expression." ([Studocu](#))

Prepare these

- Define universal gate.
- Why NAND is universal.
- Why NOR is universal.
- Implement NOT using NAND.
- Implement AND using NAND.
- Implement OR using NAND.
- Implement NOT using NOR.
- Implement AND using NOR.
- Implement OR using NOR.
- Explain XOR and XNOR as derived gates.

Priority:  **Very Important**

9. BCD to Seven-Segment Display — 8/10

Exact question

- "Draw the logic circuit diagram for BCD to seven segment conversion. Give the logic levels to display digit '3' on common anode display." ([Scribd](#))

Prepare these

- What is seven-segment display?
- Common-anode display.
- Common-cathode display.
- BCD-to-seven-segment decoder.
- Draw BCD-to-seven-segment block diagram.
- Display digits using segment logic.
- Difference between common-anode and common-cathode.

Priority:  Important

10. Encoder / Decoder — 8/10

Prepare these questions

- Define encoder.
- Define decoder.
- Explain 2-to-4 decoder.
- Explain 3-to-8 decoder.
- Explain priority encoder.
- Applications of encoder.
- Applications of decoder.
- Difference between encoder and decoder.

Priority:  Important

11. IC 7447 / IC 74138 — 7/10

Exact questions

- "State the function of IC 7447." ([Scribd](#))
- "Write a short note on IC 74138." ([Scribd](#))

Prepare these

- Function of IC 7447.
- IC 7447 as BCD-to-seven-segment decoder/driver.
- Basic purpose of IC 74138.
- Applications of IC 74138.

Priority: 🟡 Short-question preparation

12. Parity Bit — 7/10

Exact question

- "Explain the concept of parity bits. Where are parity bits used?" ([Scribd](#))

Prepare these

- Define parity bit.
- Even parity.
- Odd parity.
- Parity generation.
- Parity checking.
- Applications of parity bits.
- Limitations of parity checking.

Priority: 🟡 Medium



FINAL — TOP QUESTIONS TO PREPARE FIRST

If your objective is **maximum marks**, I would prepare in this exact order:

🔴 Tier 1 — Don't Skip

- State and prove De Morgan's theorems.
- Simplify Boolean expressions using Boolean algebra.
- Simplify Boolean expressions using K-map.
- Draw truth tables and symbols of basic logic gates.
- Explain universal gates and implement basic gates using NAND/NOR.
- Perform binary addition and subtraction.
- Perform subtraction using 2's complement.
- Convert Binary $\leftrightarrow$ Gray.
- Convert Decimal $\rightarrow$ BCD.
- Explain half adder and full adder.

- Explain half subtractor and full subtractor.
- Explain 4-bit universal adder/subtractor.
- Explain multiplexer and calculate select lines.
- Explain 1:4 / 1:8 demultiplexer.
- Convert Boolean expressions into standard SOP/POS.

● Tier 2 — Very Important

- Explain BCD-to-seven-segment decoder.
- Explain encoder and decoder.
- Explain XOR/XNOR as derived gates.
- Explain IC 7447.
- Explain IC 74138.
- Explain Gray code and its applications.
- Explain parity bits.
- Explain ASCII code.